Series of real numbers

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + \dots, \quad a_n \in \mathbb{R}, \ \forall n \geq 1$$

$(s_n)_{n \geq 1} \Rightarrow s_n = a_1 + a_2 + \dots + a_n$

$\exists \lim s_n = s \in \overline{\mathbb{R}}$

s = sum of the series

$s \in \mathbb{R} \Rightarrow$ convergent series

Th 1. (Cauchy's general test of convergence)

$\sum_{n=1}^{\infty} a_n$ is convergent $\iff \forall \varepsilon > 0, \exists n_\varepsilon \in \mathbb{N}$ s.t.

$\forall n \geq n_\varepsilon, \forall p \in \mathbb{N}$:

$$|a_{n+1} + a_{n+2} + \dots + a_{n+p}| < \varepsilon$$

Proof: $\sum_{n=1}^{\infty} a_n$ conv. $\iff s_n$ conv. $\iff (s_n)_{n \geq 1}$

fundamental $\iff \varepsilon > 0, \exists n_\varepsilon \in \mathbb{N}$ s.t. $\forall n \geq n_\varepsilon, \forall p \in \mathbb{N}$:

$$|s_{n+p} - s_n| < \varepsilon \iff |a_{n+1} + \dots + a_{n+p}| < \varepsilon, \forall n \geq n_\varepsilon$$

Th. 2. (A necessary condition of convergence) If

$\sum_{n=1}^{\infty} a_n$ is convergent $\Rightarrow \lim_{n \to \infty} a_n = 0$

Proof: $a_n = (a_1 + a_2 + \dots + a_n) - (a_1 + \dots + a_{n-1}) =$

$s_n - s_{n-1}$

$\sum_{n=1}^{\infty} a_n$ conv $\Rightarrow \exists \lim_{n\to\infty} s_n = s \in \mathbb{R}$

$\lim_{n\to\infty} a_n = \lim_{n\to\infty} (s_n - s_{n-1}) = s - s = 0$

Obs. 1) The condition $\lim a_n = 0$ is not sufficient for convergence. (Not $\Leftarrow$)

ex. $\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \ldots = +\infty$

and $\lim \frac{1}{n} = 0$

2) If $\lim a_n \neq 0 \Rightarrow \sum_{n=1}^{\infty} a_n$ is divergent

ex. $\sum_{n=1}^{\infty} \frac{n}{n+1} = +\infty$ ($\lim \frac{n}{n+1} = 1 \neq 0$)

1. Series with positive terms

$\sum_{n=1}^{\infty} a_n$, $a_n > 0$, $\forall n \geq 1$

(s_n), $n \geq 1$ is strictly increasing

Def. $\sum_{n=1}^{\infty} a_n$ is conv. $\Leftrightarrow$ (s_n), $n \geq 1$ is bounded.

Th. 1 (First comparison test). Let $\sum_{n=1}^{\infty} a_n$, $\sum_{n=1}^{\infty} b_n$ be series with positive terms s.t. $a_n \leq b_n$, $\forall n \geq n_0$.
Then: 1) If $\sum_{n=1}^{\infty} b_n$ is conv. $\Rightarrow \sum_{n=1}^{\infty} a_n$ is conv.

2) If $\sum_{n=1}^{\infty} a_n$ is div. $\Rightarrow \sum_{n=1}^{\infty} b_n$ is div.

Proof: We may suppose without loss of generality that $n_0 = 1$, because if we eliminate a finite number of terms, it's nature remains the same.

$$\left.\begin{array}{l} s_n = a_1 + a_2 + \ldots + a_n \\ t_n = b_1 + b_2 + \ldots + b_n \end{array}\right\} \overset{a_n \le b_n}{\Rightarrow} s_n \le t_n, \ \forall n \ge 1$$

1) $\sum_{n=1}^{\infty} b_n$ conv. $\Rightarrow t_n$ is convergent $\Rightarrow (t_n)$ is bounded $\Rightarrow (s_n)$ is bounded $\Rightarrow \sum_{n=1}^{\infty} a_n$ is conv.

2) $\sum_{n=1}^{\infty} a_n = +\infty \Rightarrow s_n \to \infty \Rightarrow t_n \to \infty \Rightarrow \sum_{n=1}^{\infty} b_n$ is divergent.

Ex. Test the conv. of the series:

$$\sum_{n=1}^{\infty} \frac{n}{n^3+1}, \quad a_n = \frac{n}{n^3+1}, \ n \ge 1$$

$$a_n \le \frac{n}{n^3} = \frac{1}{n^2} = b_n, \ \forall n \ge 1$$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} \text{ (convergent)} \Bigg| \Rightarrow \sum_{n=1}^{\infty} a_n \text{ conv.}$$

Th. 2. (Second comparison test) Let $\sum_{n=1}^{\infty} a_n$, $\sum_{n=1}^{\infty} b_n$ be series with positive terms s.t.:

$$\frac{a_{n+1}}{a_n} \le \frac{b_{n+1}}{b_n}, \ \forall n \ge n_0$$

Then: 1) If $\sum_{n=1}^{\infty} b_n$ is conv $\Rightarrow \sum_{n=1}^{\infty} a_n$ conv.

2) If $\sum_{n=1}^{\infty} a_n$ is div. $\Rightarrow \sum_{n=1}^{\infty} b_n$ is div.

Th. 3. (Third comparison test) Let $\sum_{n=1}^{\infty} a_n$, $\sum_{n=1}^{\infty} b_n$ be series with positive terms s.t.:

$$\exists \lim_{n\to\infty} \frac{a_n}{b_n} = \ell, \quad \ell \in [0, +\infty)$$

Then: 1) If $\ell \neq 0 \Rightarrow \sum_{n=1}^{\infty} a_n, \sum_{n=1}^{\infty} b_n$ have the same nature (both conv. / div.)

2) If $\ell = 0$ and $\sum_{n=1}^{\infty} b_n$ is div. $\Rightarrow \sum_{n=1}^{\infty} a_n$ is div.

ex. Test the conv. of the series:

$$\sum_{n=1}^{\infty} \frac{\operatorname{arctg} n}{n} = \frac{\operatorname{arctg} 1}{1} + \frac{\operatorname{arctg} 2}{2}$$

$a_n = \frac{\operatorname{arctg} n}{n}$ Compare to $\sum_{n=1}^{\infty} \frac{1}{n} = +\infty$, $b_n = \frac{1}{n}$

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \operatorname{arctg} n = \frac{\pi}{2} \neq 0 \overset{\text{Th. 3}}{\Rightarrow} \sum_{n=1}^{\infty} a_n = +\infty$$

Obs. A consequence of the comparison test is:

Let $\sum_{n=1}^{\infty} a_n$ be a series with positive terms s.t.

$\alpha \in \mathbb{R}$:

$$\lim_{n\to\infty} n^{\alpha} \cdot a_n = \ell, \quad \ell \in [0, +\infty)$$

Then: 1) If $\alpha > 1 \Rightarrow \sum_{n=1}^{\infty} a_n$ is conv.

2) If $\alpha \leq 1$ and $\ell \neq 0 \Rightarrow \sum_{n=1}^{\infty} a_n$ is div.

Proof: $\lim \frac{a_n}{\frac{1}{n^{\alpha}}} = \ell$ and compare to $\sum_{n=1}^{\infty} \frac{1}{n^{\alpha}}$

$\Rightarrow$ The conclusion is the conseq. of the third comp. test.

ex. Test the conv. of the series:

$$\sum_{n=1}^{\infty} (\sqrt{n^2+1} - n)$$

$$a_n = \sqrt{n^2+1} - n = \frac{1}{\sqrt{n^2+1} + n}$$

$$\lim n \cdot a_n = \lim \frac{n}{\sqrt{n^2+1}+n} = \frac{1}{2} \overset{L=1}{\Rightarrow} \sum_{n=1}^{\infty} a_n = +\infty$$

Th.5: (Cauchy's Root test) Let $\sum_{n=1}^{\infty} a_n$ be a series with positive terms s.t.:

$$\exists \lim \sqrt[n]{a_n} = \ell, \ \ell \in \overline{\mathbb{R}}$$

Then: 1) If $\ell < 1 \Rightarrow \sum_{n=1}^{\infty} a_n$ is convergent

2) If $\ell > 1 \Rightarrow \sum_{n=1}^{\infty} a_n$ is divergent

3) If $\ell = 1 \Rightarrow$ The test fails

ex. $\sum_{n=1}^{\infty} \left(\frac{n}{n+1}\right)^{n^2}$ convergence?

$$a_n = \left(\frac{n}{n+1}\right)^{n^2} = \left(\left(\frac{n}{n+1}\right)^{n}\right)^{n} \to 0$$

$$\lim \sqrt[n]{a_n} = \lim \left(\frac{n}{n+1}\right)^{n} = \frac{1}{e} < 1 \Rightarrow \sum_{n=1}^{\infty} a_n \text{ conv.}$$

Th.6 (D'Alembert Ratio Test) Let $\sum_{n=1}^{\infty} a_n$ be a series with positive terms s.t.:

$$\lim \frac{a_{n+1}}{a_n} = \ell, \ \ell \in \overline{\mathbb{R}}$$

Then: 1) If $\ell < 1 \Rightarrow \sum_{n=1}^{\infty} a_n$ is conv.

2) If $\ell > 1 \Rightarrow \sum_{n=1}^{\infty} a_n$ is div

3) If $\ell = 1 \Rightarrow$ Test fails

ex. $\sum_{n=1}^{\infty} \frac{\pi^n \cdot n!}{n^n}$ conv. ?

$$a_n = \frac{\pi^n \cdot n!}{n^n} \Rightarrow \lim \frac{a_{n+1}}{a_n} = \lim \frac{\pi^{n+1} \cdot (n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{\pi^n \cdot n!}$$

$$= \lim \frac{\pi \cdot (n+1)}{n+1} \cdot \left(\frac{n}{n+1}\right)^n = \frac{\pi}{e} > 1 \Rightarrow \sum_{n=1}^{\infty} a_n \text{ is div. } = +\infty$$

Th. 7: (Raabe - Duhamel Test) Let $\sum_{n=1}^{\infty} a_n$ be a series with positive terms s.t.:

$$\exists \lim n\left(\frac{a_n}{a_{n+1}} - 1\right) = \ell \in \overline{\mathbb{R}}$$

Then: 1) If $\ell < 1 \Rightarrow \sum_{n=1}^{\infty} a_n$ is divergent

2) If $\ell > 1 \Rightarrow \sum_{n=1}^{\infty} a_n$ is convergent

3) If $\ell = 1 \Rightarrow$ Test fails

ex. Study the conv. of the series:

$$\sum_{n=1}^{\infty} \frac{1 \cdot 3 \cdot 5 \cdot \ldots \cdot (2n-1)}{2 \cdot 4 \cdot 6 \cdot \ldots \cdot 2n}$$

$$a_n = \frac{1 \cdot 3 \cdot 5 \cdot \ldots \cdot (2n-1)}{2 \cdot 4 \cdot 6 \cdot \ldots \cdot 2n}$$

Ratio test: $\lim \frac{a_{n+1}}{a_n} = \frac{2n+1}{2n+2} = 1$, test fails

Raabe-Duhamel: $\lim n\left(\frac{a_n}{a_{n+1}} - 1\right) = \lim n\left(\frac{2n+2}{2n+1} - 1\right)$

$= \lim \frac{n}{2n+1} = \frac{1}{2} < 1 \Rightarrow \sum_{n=1}^{\infty} a_n$ is divergent $= +\infty$

Th. 8: (Cauchy's condensation test) Let (a_n) be a decreasing sequence of positive numbers. Then:

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + \ldots$$

and

$$\sum_{n=0}^{\infty} 2^n \cdot a_{2^n} = a_1 + 2a_2 + 4a_4 + \ldots$$

have the same nature. ($\sum_{n=1}^{\infty} a_n \sim \sum_{n=0}^{\infty} 2^n \cdot a_{2^n}$)

ex. Prove that $\sum_{n=1}^{\infty} \frac{1}{n^\alpha}$, $\alpha \in \mathbb{R}$, is conv. $\Leftrightarrow \alpha > 1$

$a_n = \frac{1}{n^\alpha}$, $\lim a_n = \begin{cases} \infty, & \alpha < 1 \\ 0, & \alpha = 0 \end{cases} \Rightarrow$

$\sum_{n=1}^{\infty} a_n$ is divergent

If $\alpha > 1$, then $a_n \searrow 0 \Rightarrow \sum_{n=1}^{\infty} a_n \sim \sum_{n=0}^{\infty} 2^n \cdot a_{2^n}$

$$\sum_{n=0}^{\infty} 2^n \cdot a_{2^n} = \sum_{n=0}^{\infty} 2^n \cdot \frac{1}{(2^n)^\alpha} = \sum_{n=0}^{\infty} (2^n)^{1-\alpha} =$$

$$\sum_{n=0}^{\infty} (2^{1-\alpha})^n = 1 + 2^{1-\alpha} + 2^{(1-\alpha)^2} + \dots \text{ (geometric series)} \Rightarrow$$

$q = 2^{1-\alpha}$, which is conv. for $q = 2^{1-\alpha} < 1 \Rightarrow \alpha > 1$

$\Rightarrow$ The GHS is convergent $\Leftrightarrow \alpha > 1$